Tirthkumar Patel

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Summary

Software Engineer with 3+ years of experience building scalable backend, full-stack, and distributed systems. Proven experience developing high-volume microservices, event-driven workflows, and cloud applications with strong reliability and observability. Skilled in Java, Spring Boot, Python, C/C++, JavaScript, Kafka, SQL, and AWS.

Professional Experience

OKX – Software Engineer, Payments Platform, *San Jose, CA, USA* *May 2024 – Present*
Tech: *Java, Spring Boot, Apache Kafka, Redis, MySQL, Prometheus, AWS, Microservices, Distributed Systems*

- Led end-to-end design and integration of payment channels across Turkey and Europe using Java, Spring Boot, Apache Kafka, and asynchronous event-driven microservices, processing 250K+ daily transactions with high throughput, fault tolerance, idempotency, distributed caching, and real-time status propagation.
- Designed and implemented a centralized payment fee platform with a self-service UI portal and dynamic rules engine across 150+ payment channels; applied Redis caching, MySQL indexing, and query optimization to reduce database load, lower fee-calculation latency, and cut manual fee setup time from 3–4 hours to under 2 minutes.
- Built reusable KYC, account verification, risk detection, and channel-eligibility components for deposit and withdrawal flows, processing 300K+ daily validations and intercepting 5–6% fraudulent or non-compliant transactions with accuracy of more than 99% through secure transactional checks and resilient service integrations.
- Integrated a fault-tolerant manual intervention platform for unknown, failed, and unresolved transactions, enabling Finance Ops to recover thousands of daily records without engineering coordination using asynchronous processing, safe retries, cached reference data, and audit trails, saving 6+ hours per day.
- Developed an internal reconciliation and monitoring service for stuck, delayed, and inconsistent payment states, using idempotent retry-safe processing, structured logging, alerting, and observability to detect 95%+ of failures.

ANB Systems LLC – Software Engineer, *Houston, TX, USA* *Jan 2024 – May 2024*
Tech: *Javascript, React.js, Node.js, MongoDB, NoSQL, Docker, AWS S3, Schema Design, Integration and End-to-End Testing*

- Designed and implemented modular full-stack features for QUICKstyle, a drag-and-drop website builder, using React.js, Node.js, and MongoDB schema design; developed reusable UI components, dynamic site-configuration workflows, and self-service publishing capabilities for 50+ clients, reducing website delivery time from one week to minutes.
- Architected a Dockerized LocalStack-based AWS integration-testing environment for Node.js services, enabling reproducible validation of REST APIs, MongoDB persistence, S3 workflows, and cloud configurations before deployment; automated end-to-end testing and debugging workflows, reducing production defect tickets by 68% and improving release reliability.

ANB Systems LLC – Machine Learning Engineer, *Houston, TX, USA* *May 2023 – August 2023*
Tech: *Python, Flask, AWS Lex, Conversational AI, NLP, Machine Learning*

- Built and deployed a conversational AI chatbot using Python, Flask, and AWS Lex, implementing intent classification, entity recognition, context handling, and fallback workflows that improved NLU accuracy by 10x.
- Optimized intent routing and response retrieval using Trie-based matching, LRU caching, semantic similarity, and confidence tuning, reducing task-completion time by 70% and improving multi-turn reliability.

DAIICT – Machine Learning Engineer, *Gandhinagar, Gujarat, India* *Jan 2022 – May 2022*
Tech: *Python, TensorFlow, Keras, OpenCV, CNN, Transfer Learning, Inception, ImageNet, Computer Vision, Deep Learning*

Deployed for campus face-mask compliance monitoring during COVID-19

- Engineered an end-to-end computer vision and deep learning pipeline for real-time face-mask detection using Python, TensorFlow, Keras, CNNs, transfer learning, an ImageNet-pretrained Inception model, OpenCV Haar Cascades, and ImageDataGenerator across 2,000+ labeled images.
- Improved face-mask classification accuracy by fine-tuning an ImageNet-pretrained Inception model with targeted data augmentation, feature extraction, and hyperparameter optimization, achieving a 2.62% gain over the baseline.
- Validated model robustness across lighting, orientation, distance, and image-quality variations using precision, recall, confusion-matrix analysis, confidence scores, and inference latency, improving reliability under real-world conditions.

Education

MS in Software Engineering Arizona State University, Tempe, AZ — Aug 2022 – May 2024
BTech in Information and Communication Technology DAIICT, India — Jul 2018 – May 2022

Technical Skills

Languages: Java, Python, C++, C, C#, Go, JavaScript, TypeScript, SQL, HTML, CSS
Frameworks: Spring Boot, Node.js, Flask, React.js, Vue.js, D3.js, Flutter
Cloud Technologies: AWS (EC2, S3, Lambda, CloudWatch), Google Cloud Platform (GCP), Azure
Databases: MySQL, PostgreSQL, MongoDB, Oracle
Tools & Technologies: GraphQL, Docker, Kubernetes, Apache Kafka, Git, Postman, Jenkins, Claude, Cursor
Machine Learning: CNN, PyTorch, OpenCV, TensorFlow, Keras
Concepts: Data Structures & Algorithms, Software & Database Schema Design, API Design, Object Oriented Programming, Distributed systems, Caching, Concurrency, Machine Learning

Projects

Custom Linux Kernel Development — *C, Linux, Operating Systems, Multithreading, Concurrency, Device Drivers*

- Built a custom operating-system kernel in C with virtual memory management, SMP multicore support, round-robin scheduling, context switching, system calls, interrupt handling, and semaphore-based synchronization, supporting 20+ concurrent threads with deadlock-free execution.
- Developed a virtual USB block-device driver with an IOCTL interface, buffer management, and kernel-space I/O handling, processing 100K+ read/write operations with zero data corruption under concurrent workloads.

Git Lazy Tracker — *Python, Flask, Vue.js, GraphQL, GitHub API, AWS*

- Developed an engineering analytics platform for managers and technical leads to identify low-engagement contributors by analyzing commit frequency, pull-request activity, code-change volume, and repository participation using Vue.js, Flask, GraphQL, and the GitHub API.
- Implemented configurable scoring algorithms for meaningful commits, contribution consistency, pull-request behavior, and inactivity patterns, reducing manual repository analysis and centralizing contributor insights.
- Designed and provisioned a serverless AWS architecture using CDK, integrating Cognito, API Gateway, Lambda, DynamoDB, S3, CloudFront, and CloudWatch to build a secure, scalable, and observable cloud application.

Dynamic Pricing Grocery Application — *Flutter, Dart, Node.js, MongoDB, Docker, Kubernetes*

- Developed a cross-platform grocery and inventory application using Flutter, Dart, Node.js, MongoDB, and REST APIs, supporting role-based authentication, product search, category filtering, QR-code lookup, cart management, inventory tracking, and administrative pricing workflows.
- Designed a dynamic pricing engine based on product expiration date, inventory levels, demand, and configurable discount constraints
- Deployed backend by containerizing a Node.js server with Docker, deploying it on a local Kubernetes cluster, and configuring GitHub Actions for automated Jest and Supertest testing.

Academic Achievement

Ranked 73rd among 250K+ students in the 12th Grade State Board Examination, placing in the top 0.03%.